

Climate Research (Excepting Observations)

- Climate research (or at least its funding) is becoming dominated more and more by large models, so called 'General Circulation Models' (GCMs) and now by 'Earth System Models' (ESMs). Some of the largest and most complex computer code ever developed.
- Similar developments are occurring in other branches of science: ecology and evolutionary biology, astrophysics, condensed matter physics.
- 'Theory', in the traditional sense, seems to be playing a smaller role. Is such 'theory' still relevant to our scientific enterprise?
- Perhaps GCMs, as the embodiment of our collective understanding, *are* our modern-day theory of climate?

This is meant to be a science talk, not just a philosophy talk, so I'll connect these questions to particular issues in climate science, for example

- ① Mixing in the ocean.
- ② Global warming.
- ③ Understanding paleo-climate regimes.